Customer Churn Prediction & Retention Analytics

Executive Summary

This project identifies likely churners, explains churn drivers, quantifies value at risk, and translates analytics findings into retention actions for a banking customer portfolio.

Metric	Value
Customers analyzed	10,000
Churn rate	20.4%
Churned customers	2,038
Annual value proxy at risk	$5,724,781
Average customer lifetime value	$12,686
High-risk customers	3,462
Tuned model ROC-AUC	0.868

Methodology

The workflow covers data understanding, cleaning, feature engineering, SQL analysis, exploratory analysis, hypothesis testing, machine learning, dashboard design, and executive storytelling. Complaint behavior is used diagnostically, while leakage-prone outcome fields are excluded from the core predictive model.

Key Insights

- Overall churn is 20.4%, representing 2,038 churned customers out of 10,000.

- Germany has the highest geography-level churn rate at 32.4%.

- The 51-60 age group has the highest age-group churn rate at 56.2%.

- The Platinum segment contributes the highest annual value proxy at risk: $2,229,843.

- The tuned retention model achieved ROC-AUC of 0.868, making it useful for campaign prioritization.

- The strongest model driver is age, indicating where retention teams should focus diagnostic review.

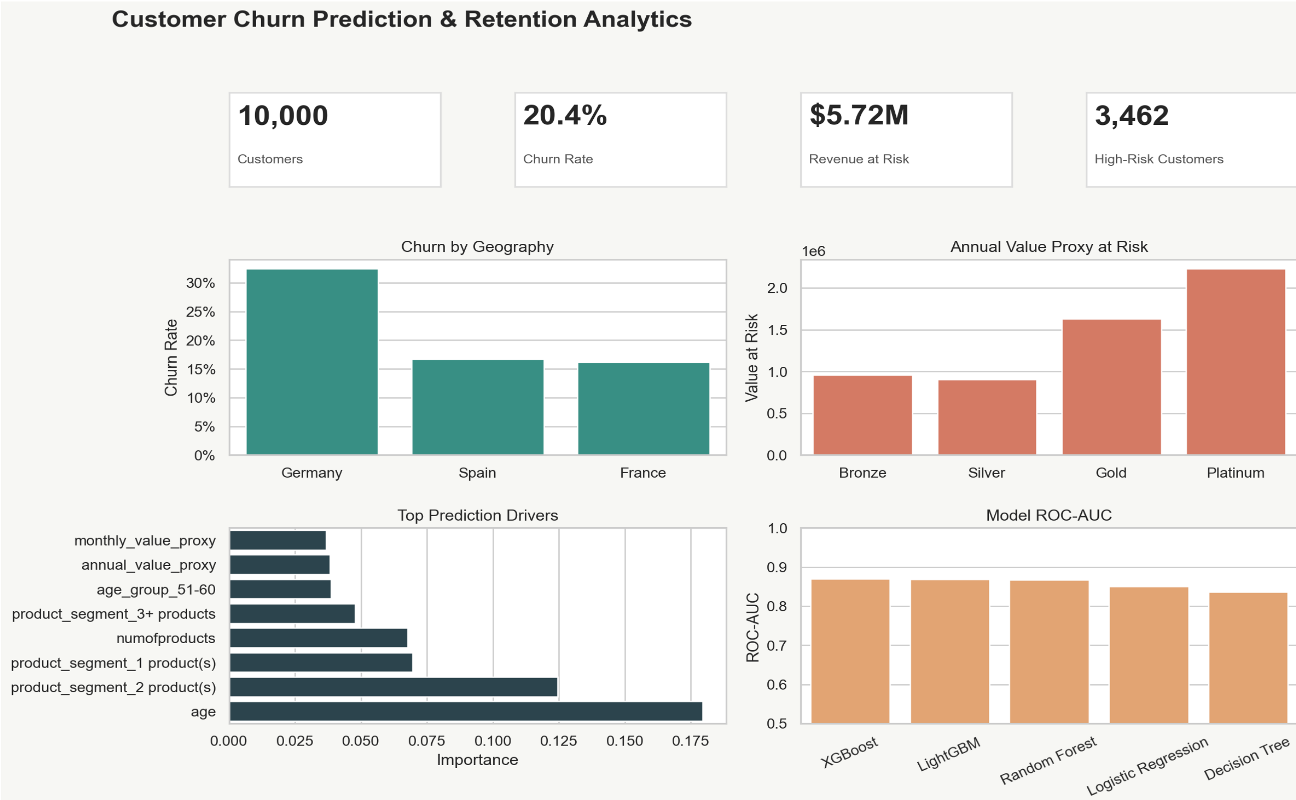
- Inactive customers and low satisfaction customers show materially higher churn risk.

Recommendations

- Launch a save campaign for high-risk, high-value customers before renewal or inactivity thresholds are reached.
- Investigate geography-specific churn drivers in Germany through service quality and product-fit analysis.
- Create age-aware retention journeys for middle-aged and older customers with complex financial needs.
- Use satisfaction and complaint signals as immediate escalation triggers for relationship managers.
- Prioritize Platinum and Gold customer segments when allocating retention budgets.
- Bundle loyalty benefits with card-type offers for customers with declining engagement.
- Track revenue at risk weekly so retention teams focus on value preservation, not only churn count.
- Refresh churn predictions monthly and route the top-risk decile into targeted outreach.
- Monitor model performance and recalibrate thresholds if churn mix changes by geography or product segment.
- Pair predictive scores with business rules so recommendations remain explainable to non-technical stakeholders.

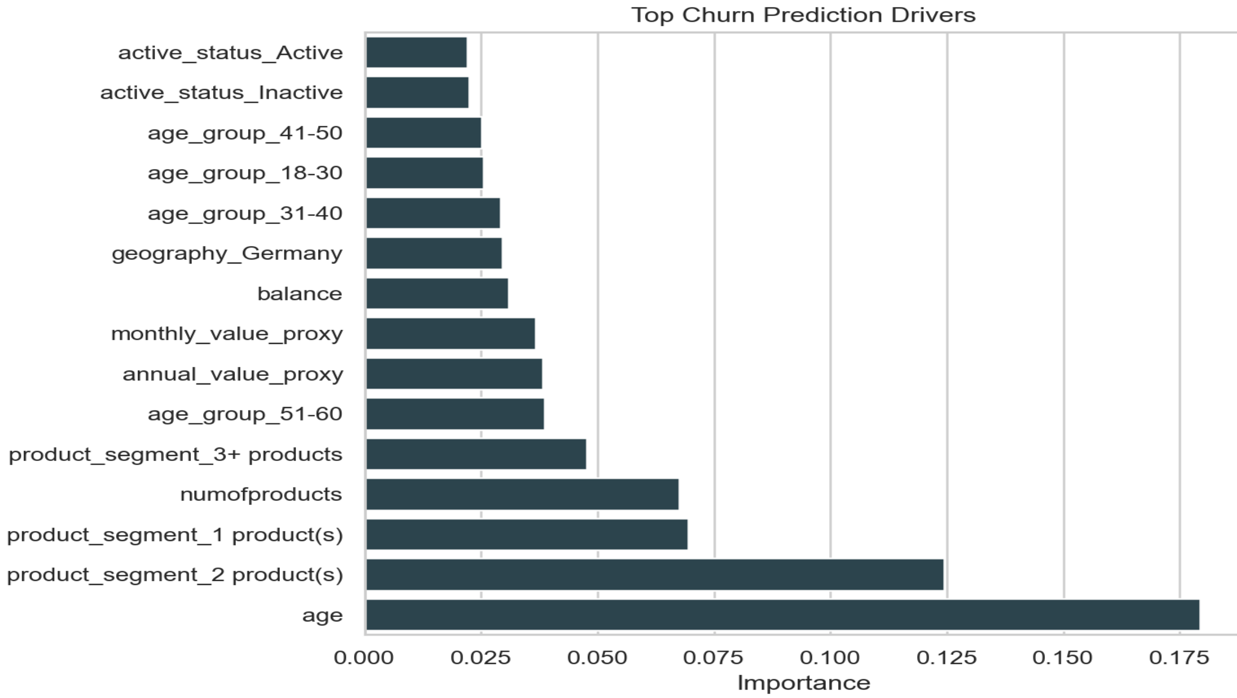
Dashboard Evidence

Dashboard Overview



Model Evidence

Feature Importance



Model Comparison